RIPGUARD: AUTONOMOUS REAL-TIME COASTAL RIP CURRENT DETECTION AND GEO-LOCALIZATION VIA EDGE-ACCELERATED SPATIO-TEMPORAL MOTION U-NET AND PLANAR HOMOGRAPHY

Principles of Computer Vision (CSE_3137)
V SEMESTER MINI PROJECT REPORT

Submitted to

MANIPAL ACADEMY OF HIGHER EDUCATION

School of Computer Engineering

Department of Computer Science & Engineering

Submitted by

Dhruv Gourisaria (245890486)
Jessica Mariam Mathews (245890380)
Samarth Bangwal (245890410)
Aadi Harnad Bhat (245890054)

Under the guidance of

Dr. Sowmya T
Assistant Professor
School of Computer Engineering
Manipal Institute of Technology, Bengaluru

MANIPAL INSTITUTE OF TECHNOLOGY
BENGALURU
(A constituent unit of MAHE, Manipal)

SCHOOL OF COMPUTER ENGINEERING
October 2026